Research Notes on Diffusion model

Diffusion model

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\, x_{t-1},\, \beta_t I\right)$$

>> Section 1

d d t φ t (x) = v t (φ t (x)) t ∈ [0 , 1] , starting from φ 0 (x) = x {\displaystyle {\frac {d}{dt}}\phi _{t}(x)=v_{t}(\phi _{t}(x))\quad t\in [0,1],\quad {\text{starting from }}\phi _{0}(x)=x}

>> Section 2

Further reading Review papers Yang, Ling (2024-09-06), YangLing0818/Diffusion-Models-Papers-Survey-Taxonomy, retrieved 2024-09-06 Yang, Ling; Zhang, Zhilong; Song, Yang; Hong, Shenda; Xu, Runsheng; Zhao, Yue; Zhang, Wentao; Cui, Bin; Yang, Ming-Hsuan (2023-11-09). "Diffusion Models: A Comprehensive Survey of Methods and Applications". ACM Comput. Surv. 56 (4): 105:1–105:39. arXiv:2209.00796. doi:10.1145/3626235. ISSN 0360-0300. Austin, Jacob; Johnson, Daniel D.; Ho, Jonathan; Tarlow, Daniel; Rianne van den Berg (2021). "Structured Denoising Diffusion Models in Discrete State-Spaces". arXiv:2107.03006 [cs.LG]. Croitoru, Florinel-Alin; Hondru, Vlad; Ionescu, Radu Tudor; Shah, Mubarak (2023-09-01). "Diffusion Models in Vision: A Survey". IEEE Transactions on Pattern Analysis and Machine Intelligence. 45 (9): 10850–10869. arXiv:2209.04747. Bibcode:2023ITPAM..4510850C. doi:10.1109/TPAMI.2023.3261988. ISSN 0162-8828. PMID 37030794. Mathematical details omitted in the article. "Power of Diffusion Models". AstraBlog. 2022-09-25. Retrieved 2023-09-25. Luo, Calvin (2022-08-25). "Understanding Diffusion Models: A Unified Perspective". arXiv:2208.11970 [cs.LG]. Weng, Lilian (2021-07-11). "What are Diffusion Models?". lilianweng.github.io. Retrieved 2023-09-25. Tutorials Nakkiran, Preetum; Bradley, Arwen; Zhou, Hattie; Advani, Madhu (2024). "Step-by-Step Diffusion: An Elementary Tutorial". arXiv:2406.08929 [cs.LG]. "Guidance: a cheat code for diffusion models". 26 May 2022. Overview of classifier guidance and classifier-free guidance, light on mathematical details. Catherine Higham, Desmond J. Higham, and Peter Grindrod: "Diffusion Models for Generative Artificial Intelligence: An Introduction for Applied Mathematicians", SIAM Review, Vol.67, No.3 (2025).

>> Section 3

L s i m p l e , t = E x 0 ∼ q ; z ∼ N (0 , I) [■ ■ θ (x t , t) − z ■ 2] {\displaystyle L_{simple,t}=E_{x_{0}\sim q;z\sim {\mathcal {N}}(0,I)}\left[\left\|\epsilon _{\theta }(x_{t},t)-z\right\|^{2}\right]}

>> Section 4

p t (x) = ∫ p t (x | z) q (z) d z and v t (x) = E q (z) [v t (x | z) p t (x | z) p t (x)] {\displaystyle p_{t}(x)=\int p_{t}(x\vert z)q(z)dz\qquad {\text{ and }}\qquad v_{t}(x)=\mathbb {E} _{q(z)}\left[{\frac {v_{t}(x\vert z)p_{t}(x\vert z)}{p_{t}(x)}}\right]}

>> Section 5

In particular, we see that we can directly sample from any point in the continuous diffusion process without going through the intermediate steps, by first sampling x 0 ∼ q , z ∼ N (0 , I) {\displaystyle x_{0}\sim q,z\sim {\mathcal {N}}(0,I)} , then get x t = e − 1 2 ∫ 0 t β (t) d t x 0 + (1 − e − ∫ 0 t β (t) d t) z {\displaystyle x_{t}=e^{-{\frac {1}{2}}\int _{0}^{t}\beta (t)dt}x_{0}+\left(1-e^{-\int _{0}^{t}\beta (t)dt}\right)z} . That is, we can quickly sample x t ∼ ρ t {\displaystyle x_{t}\sim \rho _{t}} for any t ≥ 0 {\displaystyle t\geq 0} . Now, define a certain probability distribution γ {\displaystyle \gamma } over [0 , ∞) {\displaystyle [0,\infty)} , then the score-matching loss function is defined as the expected Fisher divergence:

>> Section 6

L (θ) = E t ∼ γ , x t ∼ ρ t [■ f θ (x t , t) ■ 2 + 2 ∇ ⋅ f θ (x t , t)] {\displaystyle L(\theta)=E_{t\sim \gamma ,x_{t}\sim \rho _{t}}[\|f_{\theta }(x_{t},t)\|^{2}+2\nabla \cdot f_{\theta }(x_{t},t)]}

>> Section 7

d y t = 1 2 β (T − t) y t d t + β (T − t) ∇ y t ln ■ ρ T − t (y t) ■ score function d t + β (T − t) d W t {\displaystyle dy_{t}={\frac {1}{2}}\beta (T-t)y_{t}dt+\beta (T-t)\underbrace {\nabla _{y_{t}}\ln \rho _{T-t}\left(y_{t}\right)} _{\text{score function }}dt+{\sqrt {\beta (T-t)}}dW_{t}}

>> Section 8

Noise prediction network Since x t − 1 | x t , x 0 ∼ N (μ ~ t (x t , x 0) , σ ~ t 2 I) {\displaystyle x_{t-1}|x_{t},x_{0}\sim {\mathcal {N}}({\tilde {\mu }}_{t}(x_{t},x_{0}),{\tilde {\sigma }}_{t}^{2}I)} , this suggests that we should use μ θ (x t , t) = μ ~ t (x t , x 0) {\displaystyle \mu _{\theta }(x_{t},t)={\tilde {\mu }}_{t}(x_{t},x_{0})} ; however, the network does not have access to x 0 {\displaystyle x_{0}} , and so it has to estimate it instead. Now, since x t | x 0 ∼ N (α ¯ t x 0 , σ t 2 I) {\displaystyle x_{t}|x_{0}\sim N\left({\sqrt {{\bar {\alpha }}_{t}}}x_{0},\sigma _{t}^{2}I\right)} , we may write x t = α ¯ t x 0 + σ t z {\displaystyle x_{t}={\sqrt {{\bar {\alpha }}_{t}}}x_{0}+\sigma _{t}z} , where z {\displaystyle z} is some unknown Gaussian noise. Now we see that estimating x 0 {\displaystyle x_{0}} is equivalent to estimating z {\displaystyle z} . Therefore, let the network output a noise vector ■ θ (x t , t) {\displaystyle \epsilon _{\theta }(x_{t},t)} , and let it predict μ θ (x t , t) = μ ~ t (x t , x t − σ t ■ θ (x t , t) α ¯ t) = x t − ■ θ (x t , t) β t / σ t α t {\displaystyle \mu _{\theta }(x_{t},t)={\tilde {\mu }}_{t}\left(x_{t},{\frac {x_{t}-\sigma _{t}\epsilon _{\theta }(x_{t},t)}{\sqrt {{\bar {\alpha }}_{t}}}}\right)={\frac {x_{t}-\epsilon _{\theta }(x_{t},t)\beta _{t}/\sigma _{t}}{\sqrt {\alpha _{t}}}}} It remains to design Σ θ (x t , t) {\displaystyle \Sigma _{\theta }(x_{t},t)} . The DDPM paper suggested not learning it (since it resulted in "unstable training and poorer sample quality"), but fixing it at some value Σ θ (x t , t) = ζ t 2 I {\displaystyle \Sigma _{\theta }(x_{t},t)=\zeta _{t}^{2}I} , where either ζ t 2 = β t or σ ~ t 2 {\displaystyle \zeta _{t}^{2}=\beta _{t}{\text{ or }}{\tilde {\sigma }}_{t}^{2}} yielded similar performance. With this, the loss simplifies to L t = β t 2 2 α t σ t 2 ζ t 2 E x 0 ∼ q ; z ∼ N (0 , I) [■ ■ θ (x t , t) − z ■ 2] + C {\displaystyle L_{t}={\frac {\beta _{t}^{2}}{2\alpha _{t}\sigma _{t}^{2}\zeta _{t}^{2}}}E_{x_{0}\sim q;z\sim {\mathcal {N}}(0,I)}\left[\left\|\epsilon _{\theta }(x_{t},t)-z\right\|^{2}\right]+C} which may be minimized by stochastic gradient descent. The paper noted empirically that an even simpler loss function L s i m p l e , t = E x 0 ∼ q ; z ∼ N (0 , I) [■ ■ θ (x t , t) − z ■ 2] {\displaystyle L_{simple,t}=E_{x_{0}\sim q;z\sim {\mathcal {N}}(0,I)}\left[\left\|\epsilon _{\theta }(x_{t},t)-z\right\|^{2}\right]} resulted in better models.